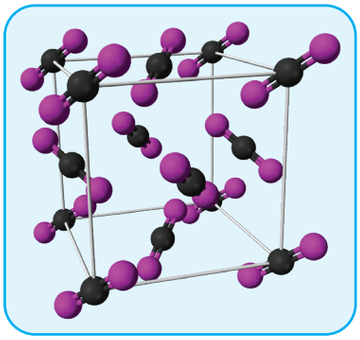
They have low melting points and are usually in liquids or gaseous state at room temperature. Examples: naphthalene, anthracene etc.,

(ii) Polar molecular solids



The constituents are molecules formed by polar covalent bonds. They are held together by relatively strong dipole-dipole interactions. They have higher melting points than the non-polar molecular solids. Examples are solid CO_2 , solid NH_3 etc.

(iii) Hydrogen bonded molecular solids

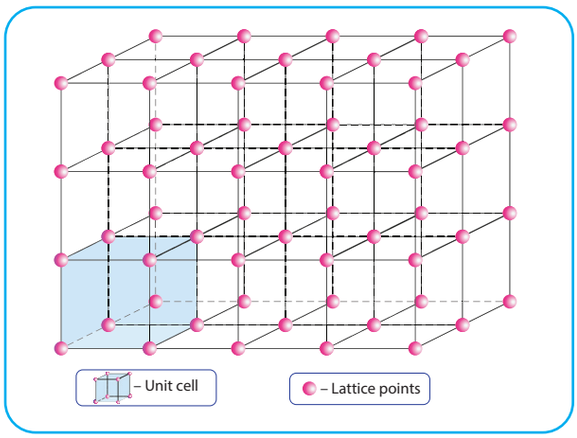
The constituents are held together by hydrogen bonds. They are generally soft solids under room temperature. Examples: solid ice (H_2O), glucose, urea etc.,

6.3.4 Metallic solids:

You have already studied in XI STD about the nature of metallic bonding. In metallic solids, the lattice points are occupied by positive metal ions and a cloud of electrons pervades the space. They are hard, and have high melting point. Metallic solids possess excellent electrical and thermal conductivity. They possess bright lustre. Examples: Metals and metal alloys belong to this type of solids, for example Cu,Fe,Zn, Ag ,Au, Cu-Zn etc.

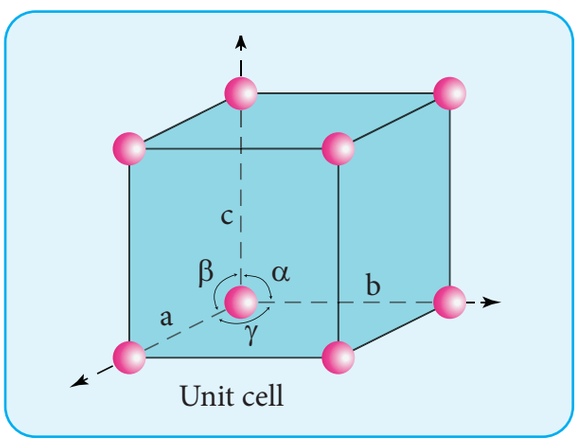
6.4 Crystal lattice and unit cell:

Crystalline solid is characterised by a definite orientation of atoms, ions or molecules, relative to one another in a three dimensional pattern. The regular arrangement of these species throughout the crystal is called a crystal lattice. A basic repeating structural unit of a crystalline solid is called a unit cell. The following figure illustrates the lattice point and the unit cell.



A crystal may be considered to consist of large number of unit cells, each one in direct contact with its nearer neighbour and all similarly oriented in space. The number of nearest neighbours that surrounding a particle in a crystal is called the coordination number of that particle.

A unit cell is characterised by the three edge lengths or lattice constants a ,b and c and the angle between the edges α, β and γ



XII U6 Solid State - Jerald.indd 180 04-01-2021 13:28:01